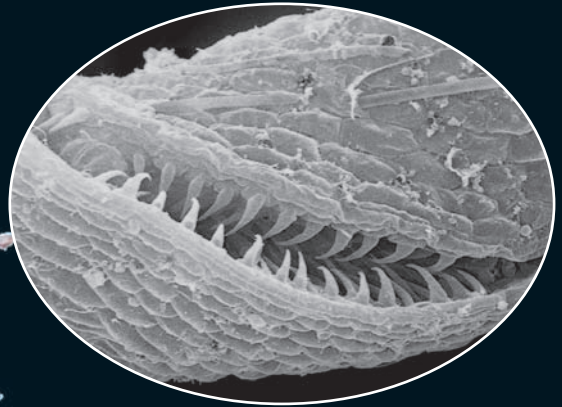


37

Soil and Plant Nutrition



KEY CONCEPTS

- 37.1** Soil contains a living, complex ecosystem
- 37.2** Plants require essential elements to complete their life cycle
- 37.3** Plant nutrition often involves relationships with other organisms

▲ **Figure 37.1** Does this plant have roots?

The Corkscrew Carnivore

The pale, rootlike appendages of *Genlisea*, the wetland herb seen in **Figure 37.1**, are actually highly modified underground leaves adapted for trapping and digesting a variety of small soil inhabitants, including bacteria, algae, protozoa, nematodes, and copepods. But how do these trap-leaves work? Imagine twisting a narrow strip of paper to make a drinking straw. This is essentially the mechanism by which these corkscrew-shaped tubular leaves form. A narrow spiral slit runs along most of the trap-leaf's length; it is lined with curved hairs that allow microorganisms to enter the leaf tube but not exit. Once inside, prey find themselves traveling inexorably upward toward a small chamber lined with digestive glands that seal their fate. The inability of prey to backtrack is ensured by another set of curved hairs that allow only one-way passage (see micrograph at left). *Genlisea*'s carnivorous habit is a marvelous adaptation that enables the plant to supplement the meager mineral rations available from the boggy, nutrient-poor soils in which it grows with minerals released from its digested prey.

As discussed in Chapter 36, plants obtain nutrients from both the atmosphere and the soil. Using sunlight as an energy source, they produce organic nutrients by reducing carbon dioxide to sugars through the process of photosynthesis. They also take up water and various inorganic nutrients from the soil through their root systems. This chapter focuses on plant nutrition, the study of the minerals necessary



for plant growth. After discussing the physical properties of soils and factors that govern soil quality, we'll explore why certain mineral nutrients are essential for plant function. Finally, we examine some nutritional adaptations that have evolved, often in relationships with other organisms.

CONCEPT 37.1

Soil contains a living, complex ecosystem

The upper layers of the soil, from which plants absorb nearly all of the water and minerals they require, contain a wide range of living organisms that interact with each other and with the physical environment. This complex ecosystem may take centuries to form but can be destroyed by human mismanagement in just a few years. To understand why soil must be conserved and why particular plants grow where they do, it is necessary to first consider the basic physical properties of soil: its texture and composition.

Soil Texture

The texture of soil depends on the sizes of its particles. Soil particles can range from coarse sand (0.02–2 mm in diameter) to silt (0.002–0.02 mm) to microscopic clay particles (less than 0.002 mm). These different-sized particles arise ultimately from the weathering of rock. Water freezing in crevices of rocks causes mechanical fracturing, and weak acids in the soil break rocks down chemically. When organisms penetrate the rock, they accelerate breakdown by chemical and mechanical means. Roots, for example, secrete acids that dissolve the rock, and their growth in fissures leads to mechanical fracturing. Mineral particles released by weathering become mixed with living organisms and **humus**, the remains of dead organisms and other organic matter, forming **topsoil**. The topsoil and other soil layers are called **soil horizons** (Figure 37.2). The topsoil, or A horizon, can range in depth from millimeters to meters. We focus mostly on properties of topsoil because it is generally the most important soil layer for plant growth.

In the topsoil, plants are nourished by the soil solution, the water and dissolved minerals in the pores between soil particles. The pores also contain air pockets. After a heavy rain, water drains from the larger spaces in the soil, but smaller spaces retain water because water molecules are attracted to the negatively charged surfaces of clay and other particles.

The topsoils that are the most fertile—supporting the most abundant growth—are **loams**, which are composed of roughly equal amounts of sand, silt, and clay. Loamy soils have enough small silt and clay particles to provide ample surface area for the adhesion and retention of minerals and water. Meanwhile, the large spaces between sand particles enable efficient diffusion of oxygen to the roots. Sandy soils



The A horizon is the topsoil, a mixture of broken-down rock of various textures, living organisms, and decaying organic matter.

The B horizon contains much less organic matter than the A horizon and is less weathered.

The C horizon is composed mainly of partially broken-down rock. Some of the rock served as "parent" material for minerals that later helped form the upper horizons.

▲ Figure 37.2 Soil horizons.

generally don't retain enough water to support vigorous plant growth, and clayey soils tend to retain too much water. When soil does not drain adequately, the air is replaced by water, and the roots suffocate from lack of oxygen. Typically, the most fertile topsoils have pores that are about half water and half air, providing a good balance between aeration, drainage, and water storage capacity. The physical properties of soils can be adjusted by adding soil amendments, such as peat moss, compost, manure, or sand.

Topsoil Composition

A soil's composition encompasses its inorganic (mineral) and organic chemical components. The organic components include the many life-forms that inhabit the soil.

Inorganic Components

The surface charges of soil particles determine their ability to bind many nutrients. Most soil particles are negatively charged. Positively charged ions (cations)—such as potassium (K^+), calcium (Ca^{2+}), and magnesium (Mg^{2+})—adhere to these particles and are less easily lost by *leaching*, the percolation of water through the soil.

Roots, however, do not absorb mineral cations directly from soil particles; they absorb them from the soil solution. Mineral cations enter the soil solution by **cation exchange**, a process in which cations are displaced from soil particles by other cations, particularly H^+ (Figure 37.3). Therefore, a soil's capacity to exchange cations is determined by the number of cation adhesion sites and by the soil's pH. In general, the more clay and organic matter in the soil, the higher the cation exchange capacity. The clay content is important

Soil Conservation and Sustainable Agriculture

Ancient farmers recognized that yields on a particular plot of land decreased over the years. Moving to uncultivated areas, they observed the same pattern of reduced yields over time. Eventually, they realized that fertilization could make soil a renewable resource that enabled crops to be cultivated season after season at a fixed location. This sedentary agriculture facilitated a new way of life. Humans began to build permanent dwellings—the first villages. They also stored food for use between harvests, and food surpluses enabled some people to specialize in nonfarming occupations. In short, soil management, by fertilization and other practices, helped prepare the way for modern societies.

Unfortunately, soil mismanagement has been a recurrent problem throughout human history, as exemplified by the American Dust Bowl, an ecological and human disaster that ravaged the southwestern Great Plains of the United States in the 1930s. This region suffered through devastating dust storms that resulted from a prolonged drought and decades of inappropriate farming techniques. Before the arrival of farmers, the Great Plains had been covered by hardy grasses that held the soil in place in spite of recurring droughts and torrential rains. But in the late 1800s and early 1900s, many homesteaders settled in the region, planting wheat and raising cattle. These land uses left the soil exposed to erosion by winds. A few years of drought made the problem worse. During the 1930s, huge quantities of fertile soil were blown away in “black blizzards,” rendering millions of hectares of farmland useless (**Figure 37.4**). In one of the worst dust storms, clouds of dust blew eastward to Chicago, where soil fell like snow, and even reached the Atlantic coast. Hundreds of thousands of people in the Dust Bowl region were forced to abandon their homes and land, a plight immortalized in John Steinbeck’s novel *The Grapes of Wrath*.

Soil mismanagement continues to be a major problem to this day. More than 30% of the world’s farmland has reduced productivity stemming from poor soil conditions, such as chemical contamination, mineral deficiencies, acidity,

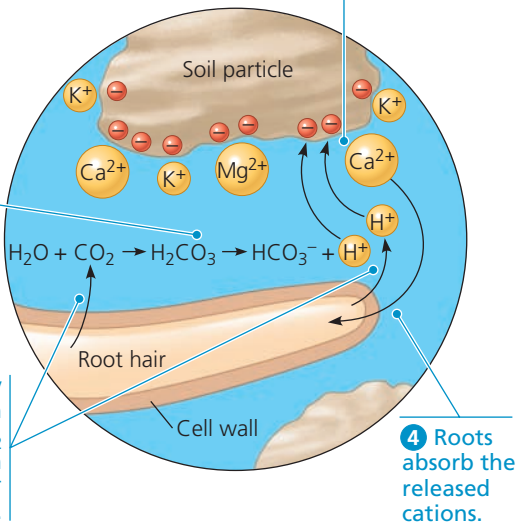


▲ **Figure 37.4** A massive dust storm in the American Dust Bowl during the 1930s.

3 H⁺ ions in the soil solution neutralize the negative charge of soil particles, causing release of mineral cations into the soil solution.

2 CO₂ reacts with H₂O to form H₂CO₃, which releases H⁺ upon disassociation.

1 Roots acidify the soil solution by releasing CO₂ from respiration and pumping H⁺ into the soil.



4 Roots absorb the released cations.

▲ **Figure 37.3** Cation exchange in soil.

? Which are more likely to be leached from the soil by heavy rains—cations or anions? Explain.

because these small particles have a high ratio of surface area to volume. Negatively charged ions (anions)—such as the plant nutrients nitrate (NO₃[−]), phosphate (H₂PO₄[−]), and sulfate (SO₄^{2−})—do not bind to the negatively charged soil particles typically found in the most productive soils and are therefore more easily lost by leaching.

Organic Components

The major organic component of topsoil is humus, which consists of organic material produced by the decomposition of fallen leaves, dead organisms, feces, and other organic matter by bacteria and fungi. Humus prevents clay particles from packing together and forms a crumbly soil that retains water but is still porous enough to aerate roots. Humus also increases the soil’s capacity to exchange cations and is a reservoir of mineral nutrients that return gradually to the soil as microorganisms decompose the organic matter.

Topsoil is home to an astonishing number and variety of organisms. A teaspoon of topsoil has about 5 billion bacteria, which cohabit with fungi, algae and other protists, insects, earthworms, nematodes, and plant roots. The activities of all these organisms affect the soil’s physical and chemical properties. Earthworms, for example, consume organic matter and derive their nutrition from the bacteria and fungi growing on this material. They excrete wastes and move large amounts of material to the soil surface. In addition, they move organic matter into deeper layers. Earthworms mix and clump the soil particles, allowing for better gaseous diffusion and water retention. Roots also affect soil texture and composition. For example, by binding the soil, they reduce erosion, and by excreting acids, they lower soil pH.



▲ **Figure 37.5 Sudden land subsidence.** Overuse of groundwater for irrigation triggered formation of this sinkhole in Florida.

salinity, and poor drainage. As the world's population grows, the demand for food increases. Because soil quality greatly affects crop yield, soil resources must be managed prudently.

We'll now discuss how farmers irrigate and modify soil in order to maintain good crop yields. The goal is **sustainable agriculture**, a commitment embracing a variety of farming methods that are conservation minded, environmentally safe, and profitable. We will also examine problems and solutions relating to soil degradation.

Irrigation

Because water is often the limiting factor in plant growth, perhaps no technology has increased crop yield as much as irrigation. However, irrigation is a huge drain on freshwater resources. Globally, about 75% of all freshwater use is devoted to agriculture. Many rivers in arid regions have been reduced to trickles by the diversion of water for irrigation. The primary source of irrigation water, however, is not surface waters, such as rivers and lakes, but underground water reserves called *aquifers*. In some parts of the world, the rate of water removal is exceeding the natural refilling of the aquifers. The result is *land subsidence*, a gradual settling or sudden sinking of Earth's surface (**Figure 37.5**). Land subsidence alters drainage patterns, causes damage to human-made structures, contributes to loss of underground springs, and increases the risk of flooding.

Irrigation, particularly from groundwater, can also lead to soil *salinization*—the addition of salts to the soil that make it too salty for cultivating plants. Salts dissolved in irrigation water accumulate in the soil as the water evaporates, making the water potential of the soil solution more negative. The water potential gradient from soil to roots is reduced, diminishing water uptake (see Chapter 36).

Many forms of irrigation, such as the flooding of fields, are wasteful because much of the water evaporates. To use water efficiently, farmers must understand the water-holding capacity of their soil, the water needs of their crops, and the

appropriate irrigation technology. One popular technology is *drip irrigation*, the slow release of water to soil and plants from perforated plastic tubing placed directly at the root zone. Because drip irrigation requires less water and reduces salinization, it is used in many arid agricultural regions.

Fertilization

In natural ecosystems, mineral nutrients are usually recycled by the excretion of animal wastes and the decomposition of humus. Agriculture, however, is unnatural. The lettuce you eat, for example, contains minerals extracted from a farmer's field. As you excrete wastes, these minerals are deposited far from their original source. Over many harvests, the farmer's field will eventually become depleted of nutrients. Nutrient depletion is a major cause of global soil degradation. Farmers must reverse nutrient depletion by means of **fertilization**, the addition of mineral nutrients to the soil.

Today, most farmers in industrialized nations use fertilizers containing minerals that are either mined or prepared by energy-intensive processes. These fertilizers are usually enriched in nitrogen (N), phosphorus (P), and potassium (K)—the nutrients most commonly deficient in depleted soils. You may have seen fertilizers labeled with a three-number code, called the N–P–K ratio. A fertilizer marked “15–10–5,” for instance, is 15% N (as ammonium or nitrate), 10% P (as phosphate), and 5% K (as the mineral potash).

Manure, fishmeal, and compost are called “organic” fertilizers because they are of biological origin and contain decomposing organic material. Before plants can use organic material, however, it must be decomposed into the inorganic nutrients that roots can absorb. Whether from organic fertilizer or a chemical factory, the minerals a plant extracts are in the same form. However, organic fertilizers release them gradually, whereas minerals in commercial fertilizers are immediately available but may not be retained by the soil for long. Minerals not absorbed by roots are often leached from the soil by rainwater or irrigation. To make matters worse, mineral runoff into lakes may lead to explosions in algal populations that can deplete oxygen levels and decimate fish populations.

Adjusting Soil pH

Soil pH is an important factor that influences mineral availability by its effect on cation exchange and the chemical form of minerals. Depending on the soil pH, a particular mineral may be bound too tightly to clay particles or may be in a chemical form that the plant cannot absorb. Most plants prefer slightly acidic soil because the high H^+ concentrations can displace positively charged minerals from soil particles, making them more available for absorption. Adjusting soil pH is tricky because a change in H^+ concentration may make one mineral more available but another less available. At pH 8, for instance, plants can absorb calcium, but iron

is almost unavailable. The soil pH should be matched to a crop's mineral needs. If the soil is too alkaline, adding sulfate will lower the pH. Soil that is too acidic can be adjusted by adding lime (calcium carbonate or calcium hydroxide).

When the soil pH dips to 5 or lower, toxic aluminum ions (Al^{3+}) become more soluble and are absorbed by roots, stunting root growth and preventing the uptake of calcium, a needed plant nutrient. Some plants can cope with high Al^{3+} levels by secreting organic anions that bind Al^{3+} and render it harmless. However, low soil pH and Al^{3+} toxicity continue to pose serious problems, especially in tropical regions, where the pressure of producing food for a growing population is often most acute.

Controlling Erosion

As happened most dramatically in the Dust Bowl, water and wind erosion can remove large amounts of topsoil. Erosion is a major cause of soil degradation because nutrients are carried away by wind and streams. To limit erosion, farmers plant rows of trees as windbreaks, terrace hillside crops, and cultivate crops in a contour pattern (**Figure 37.6**). Crops such as alfalfa and wheat provide good ground cover and protect the soil better than maize and other crops that are usually planted in more widely spaced rows.

Erosion can also be reduced by a plowing technique called **no-till agriculture**. In traditional plowing, the entire field is tilled, or turned over. This practice helps control weeds but disrupts the meshwork of roots that holds the soil in place, leading to increased surface runoff and erosion. In no-till agriculture, a special plow creates narrow furrows for seeds and fertilizer. In this way, the field is seeded with minimal disturbance to the soil, while also using less fertilizer.

Phytoremediation

Some land areas are unfit for cultivation because toxic metals or organic pollutants have contaminated the soil or groundwater. Traditionally, soil remediation, the detoxification of contaminated soils, has focused on nonbiological technologies, such as removing and storing contaminated soil in landfills, but these techniques are costly and often disrupt the landscape. **Phytoremediation** is a nondestructive biotechnology that harnesses the ability of some plants to extract soil pollutants and concentrate them in portions of the plant that can be easily removed for safe disposal. For example, alpine pennycress (*Thlaspi caerulescens*) can accumulate zinc in its shoots at concentrations 300 times higher than most plants can tolerate. The shoots can be harvested and the zinc removed. Such plants show promise for cleaning up areas contaminated by smelters, mines, or nuclear tests. Phytoremediation is a type of bioremediation, which also uses prokaryotes and protists to detoxify polluted sites (see Chapters 27 and 55).



▲ **Figure 37.6 Contour tillage.** These crops are planted in rows that go around, rather than up and down, the hills. Contour tillage helps slow water runoff and topsoil erosion after heavy rains.

We have discussed the importance of soil conservation for sustainable agriculture. Mineral nutrients contribute greatly to soil fertility, but which minerals are most important, and why do plants need them? These are the topics of the next section.

CONCEPT CHECK 37.1

1. Explain how the phrase “too much of a good thing” can apply to watering and fertilizing plants.
2. Some lawn mowers collect clippings. What is a drawback of this practice with respect to plant nutrition?
3. **WHAT IF?** How would adding clay to loamy soil affect capacity to exchange cations and retain water? Explain.
4. **MAKE CONNECTIONS** Note three ways the properties of water contribute to soil formation. See Concept 3.2.

For suggested answers, see Appendix A.

CONCEPT 37.2

Plants require essential elements to complete their life cycle

Water, air, and soil minerals all contribute to plant growth. A plant's water content can be measured by comparing the mass before and after drying. Typically, 80–90% of a plant's fresh mass is water. Some 96% of the remaining dry mass consists of carbohydrates such as cellulose and starch that are produced by photosynthesis. Thus, the components of carbohydrates—carbon, oxygen, and hydrogen—are the most abundant elements in dried plant residue. Inorganic substances from the soil, although essential for plant survival, account for only about 4% of a plant's dry mass.

Essential Elements

The inorganic substances in plants contain more than 50 chemical elements. In studying the chemical composition of plants, we must distinguish elements that are essential from those that are merely present in the plant. A chemical element is considered an **essential element** only if it is required for a plant to complete its life cycle and produce another generation.

To determine which chemical elements are essential, researchers use **hydroponic culture**, in which plants are grown in mineral solutions instead of soil (**Figure 37.7**). Such studies have helped identify 17 essential elements needed by all plants (**Table 37.1**). Hydroponic culture is also used on a small scale to grow some greenhouse crops.

Nine of the essential elements are called **macronutrients** because plants require them in relatively large amounts. Six of these are the major components of organic compounds forming a plant's structure: carbon, oxygen, hydrogen, nitrogen, phosphorus, and sulfur. The other three macronutrients are potassium, calcium, and magnesium. Of all the mineral nutrients, nitrogen contributes the most to plant growth and crop yields.

The other essential elements are called **micronutrients** because plants need them in only tiny quantities. They are chlorine, iron, manganese, boron, zinc, copper, nickel, and molybdenum. In some cases, sodium may be a ninth essential micronutrient: Plants that use the C_4 and CAM pathways of photosynthesis (see Chapter 10) require sodium ions to regenerate phosphoenolpyruvate, which is the CO_2 acceptor in these two types of carbon fixation.

Micronutrients function in plants mainly as cofactors, nonprotein helpers in enzymatic reactions (see Chapter 8). Iron, for example, is a metallic component of cytochromes, the proteins in the electron transport chains of chloroplasts

and mitochondria. It is because micronutrients generally play catalytic roles that plants need only tiny quantities. The requirement for molybdenum, for instance, is so modest that there is only one atom of this rare element for every 60 million atoms of hydrogen in dried plant material. Yet a deficiency of molybdenum or any other micronutrient can weaken or kill a plant.

Symptoms of Mineral Deficiency

The symptoms of a deficiency depend partly on the mineral's function as a nutrient. For example, a deficiency of magnesium, a component of chlorophyll, causes *chlorosis*, yellowing of leaves. In some cases, the relationship between a deficiency and its symptoms is less direct. For instance, iron deficiency can cause chlorosis even though chlorophyll contains no iron, because iron ions are required as a cofactor in an enzymatic step of chlorophyll synthesis.

Mineral deficiency symptoms depend not only on the role of the nutrient but also on its mobility within the plant. If a nutrient moves about freely, symptoms appear first in older organs because young, growing tissues are a greater sink for nutrients that are in short supply. For example, magnesium is relatively mobile and is shunted preferentially to young leaves. Therefore, a plant deficient in magnesium first shows signs of chlorosis in its older leaves. The mechanism for preferential routing is the source-to-sink translocation in phloem, as minerals move along with sugars to the growing tissues (see Figure 36.16). In contrast, a deficiency of a mineral that is relatively immobile affects young parts of the plant first. Older tissues may have adequate amounts that they retain during periods of short supply. For example, iron does not move freely within a plant, and an iron deficiency causes yellowing of young leaves before any effect on older leaves is visible. The mineral requirements of a plant may also change with the time of the year and the age of the plant. Young seedlings, for example, rarely show mineral deficiency symptoms because their mineral requirements are met largely by minerals released from stored reserves in the seed itself.

Deficiencies of phosphorus, potassium, and especially nitrogen are most common. Micronutrient shortages are less common and tend to occur in certain geographic regions because of differences in soil composition. Symptoms of a deficiency may vary between species but are often distinctive enough for a plant physiologist or farmer to diagnose (see the **Scientific Skills Exercise**). One way to confirm a diagnosis is to analyze the mineral content of the plant or soil. The amount of a micronutrient needed to correct a deficiency is usually small. For example, a zinc deficiency in fruit trees can usually be cured by hammering a few zinc nails into each tree trunk. Moderation is important because overdoses can be detrimental or toxic. Too much nitrogen, for example, can lead to excessive vine growth in tomato plants at the expense of good fruit production.

▼ Figure 37.7 Research Method

Hydroponic Culture

Application In hydroponic culture, plants are grown in mineral solutions without soil. One use of hydroponic culture is to identify essential elements in plants.

Technique Plant roots are bathed in aerated solutions of known mineral composition. Aerating the water provides the roots with oxygen for cellular respiration. (Note: The flasks would normally be opaque to prevent algal growth.) A mineral, such as potassium, can be omitted to test whether it is essential.



Results If the omitted mineral is essential, mineral deficiency symptoms occur, such as stunted growth and discolored leaves. By definition, the plant would not be able to complete its life cycle. Deficiencies of different elements may have different symptoms, which can aid in diagnosing mineral deficiencies in soil.

Table 37.1 Essential Elements in Plants

Element (Form Primarily Absorbed by Plants)	% Mass in Dry Tissue	Major Functions	Early Visual Symptoms of Nutrient Deficiencies
Macronutrients			
Carbon (CO ₂)	45%	Major component of plant's organic compounds	Poor growth
Oxygen (CO ₂)	45%	Major component of plant's organic compounds	Poor growth
Hydrogen (H ₂ O)	6%	Major component of plant's organic compounds	Wilting, poor growth
Nitrogen (NO ₃ ⁻ , NH ₄ ⁺)	1.5%	Component of nucleic acids, proteins, and chlorophyll	Chlorosis at tips of older leaves (common in heavily cultivated soils or soils low in organic material)
Potassium (K ⁺)	1.0%	Cofactor of many enzymes; major solute functioning in water balance; operation of stomata	Mottling of older leaves, with drying of leaf edges; weak stems; roots poorly developed (common in acidic or sandy soils)
Calcium (Ca ²⁺)	0.5%	Important component of middle lamella and cell walls; maintains membrane function; signal transduction	Crinkling of young leaves; death of terminal buds (common in acidic or sandy soils)
Magnesium (Mg ²⁺)	0.2%	Component of chlorophyll; cofactor of many enzymes	Chlorosis between veins, found in older leaves (common in acidic or sandy soils)
Phosphorus (H ₂ PO ₄ ⁻ , HPO ₄ ²⁻)	0.2%	Component of nucleic acids, phospholipids, ATP	Healthy appearance but very slow development; thin stems; purpling of veins; poor flowering and fruiting (common in acidic, wet, or cold soils)
Sulfur (SO ₄ ²⁻)	0.1%	Component of proteins	General chlorosis in young leaves (common in sandy or very wet soils)
Micronutrients			
Chlorine (Cl ⁻)	0.01%	Photosynthesis (water-splitting); functions in water balance	Wilting; stubby roots; leaf mottling (uncommon)
Iron (Fe ³⁺ , Fe ²⁺)	0.01%	Respiration; photosynthesis: chlorophyll synthesis; N ₂ fixation	Chlorosis between veins, found in young leaves (common in basic soils)
Manganese (Mn ²⁺)	0.005%	Active in formation of amino acids; activates some enzymes; required for water-splitting step of photosynthesis	Chlorosis between veins, found in young leaves (common in basic soils rich in humus)
Boron (H ₂ BO ₃ ⁻)	0.002%	Cofactor in chlorophyll synthesis; role in cell wall function; pollen tube growth	Death of meristems; thick, leathery, and discolored leaves (occurs in any soil; most common micronutrient deficiency)
Zinc (Zn ²⁺)	0.002%	Active in formation of chlorophyll; cofactor of some enzymes; needed for DNA transcription	Reduced internode length; crinkled leaves (common in some geographic regions)
Copper (Cu ⁺ , Cu ²⁺)	0.001%	Component of many redox and lignin-biosynthetic enzymes	Light green color throughout young leaves, with drying of leaf tips; roots stunted and excessively branched (common in some geographic regions)
Nickel (Ni ²⁺)	0.001%	Nitrogen metabolism	General chlorosis in all leaves; death of leaf tips (common in acidic or sandy soils)
Molybdenum (MoO ₄ ²⁻)	0.0001%	Nitrogen metabolism	Death of root and shoot tips; chlorosis in older leaves (common in acidic soils in some geographic areas)

MAKE CONNECTIONS Explain why CO₂, rather than O₂, is the source of much of the dry mass oxygen in plants. See Concept 10.1.

Improving Plant Nutrition by Genetic Modification

In exploring plant nutrition so far, we have discussed how farmers use irrigation, fertilization, and other means to tailor soil conditions for a crop. An opposite approach is tailoring the plant by genetic engineering to better fit the soil. Here we highlight two examples of how genetic engineering improves plant nutrition and fertilizer usage.

Resistance to Aluminum Toxicity

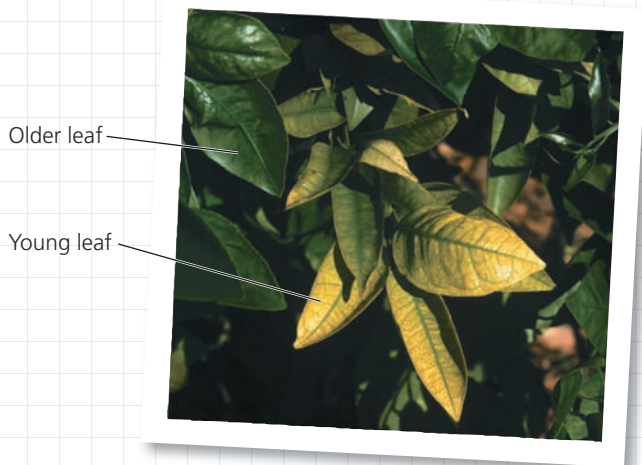
Aluminum in acidic soils damages roots and reduces crop yields. The major mechanism of aluminum resistance is secretion of organic acids (such as malic acid and citric acid) by roots. These acids bind to free aluminum ions and lower the levels of aluminum in the soil. Scientists have altered tobacco and papaya plants by introducing a citrate synthase gene from a bacterium into the plants' genomes. The resulting overproduction of citric acid increased aluminum resistance.

SCIENTIFIC SKILLS EXERCISE

Making Observations

What Mineral Deficiency Is This Plant Exhibiting? Plant growers often diagnose deficiencies in their crops by examining changes to the foliage, such as chlorosis (yellowing), death of some leaves, discoloring, mottling, scorching, or changes in size or texture. In this exercise, you will diagnose a mineral deficiency by observing a plant's leaves and applying what you have learned about symptoms from the text and Table 37.1.

Data The data for this exercise come from the photograph below of leaves on an orange tree exhibiting a mineral deficiency.



Interpret the Data

1. How do the young leaves differ in appearance from the older leaves?
2. In three words, what is the most prominent mineral deficiency symptom seen in this photo? List the three nutrients whose deficiencies give rise to this symptom. Based on the symptom's location, which one of these three nutrients can be ruled out, and why? What does the location suggest about the other two nutrients?
3. How would your hypothesis about the cause of this deficiency be influenced if tests showed that the soil was low in humus?

 A version of this Scientific Skills Exercise can be assigned in MasteringBiology.

Smart Plants

Agricultural researchers are developing ways to maintain crop yields while reducing fertilizer use. One approach is to genetically engineer “smart” plants that signal when a nutrient deficiency is imminent—but *before* damage has occurred. One type of smart plant takes advantage of a promoter (a DNA sequence indicating where the transcription of a gene starts) that more readily binds RNA polymerase (the transcription enzyme) when the phosphorus content of the plant's tissues begins to decline. This promoter is linked to a “reporter” gene that leads to production of a light blue pigment in the leaf cells (**Figure 37.8**). When leaves of these smart plants develop a blue tinge, the farmer knows it is time to add phosphate-containing fertilizer.



No phosphorus deficiency



Beginning phosphorus deficiency



Well-developed phosphorus deficiency

▲ **Figure 37.8** Deficiency warnings from “smart” plants. Some plants have been genetically modified to signal an impending nutrient deficiency before irreparable damage occurs. For example, after laboratory treatments, the research plant *Arabidopsis* develops a blue color in response to an imminent phosphate deficiency.

So far, you have learned that soil, to support vigorous plant growth, must have an adequate supply of mineral nutrients, sufficient aeration, good water-holding capacity, low salinity, and a pH near neutrality. It must also be free of toxic concentrations of minerals and other chemicals. These physical and chemical features of soil, however, are just part of the story: We must also consider the living components of soil.

CONCEPT CHECK 37.2

1. Are some essential elements more important than others? Explain.
2. **WHAT IF?** If an element increases the growth rate of a plant, can it be defined as an essential element?
3. **MAKE CONNECTIONS** Based on the information on fermentation in Figure 9.17, explain why hydroponically grown plants would grow much more slowly if they were not sufficiently aerated.

For suggested answers, see Appendix A.

CONCEPT 37.3

Plant nutrition often involves relationships with other organisms

To this point, we have portrayed plants as exploiters of soil resources. However, plants and soil actually have a two-way relationship. Dead plants provide much of the energy that is needed by soil-dwelling microorganisms, while the secretions produced by living roots support a wide variety of microbes. Here we'll focus on some of the *mutualistic*—mutually beneficial—relationships between plants and bacteria or fungi. Then we'll look at some unusual plants that form nonmutualistic relationships with other plants or, in a few cases, with animals.

Bacteria and Plant Nutrition

A variety of mutualistic bacteria play roles in plant nutrition.

Rhizobacteria live in the **rhizosphere**, the soil closely surrounding the plant's roots. **Endophytes** are nonpathogenic bacteria (or fungi) that live between cells within the plant itself but do not form deep, intimate associations with the cells or alter their morphology. Both endophytic bacteria and rhizobacteria depend on nutrients such as sugars, amino acids, and organic acids that are secreted by plant cells. In the case of the rhizosphere, up to 20% of a plant's photosynthetic production fuels the organisms in this miniature ecosystem. In turn, endophytic bacteria and rhizobacteria enhance plant growth by a variety of mechanisms. Some produce chemicals that stimulate plant growth. Others produce antibiotics that protect roots from disease. Still

others absorb toxic metals or make nutrients more available to roots. Inoculation of seeds with plant-growth-promoting rhizobacteria can increase crop yield and reduce the need for fertilizers and pesticides.

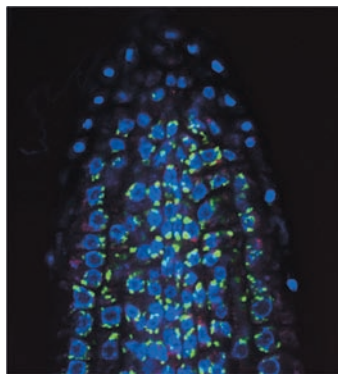
Both the intercellular spaces occupied by endophytic bacteria and the rhizosphere associated with each plant root system contain a unique and complex cocktail of root secretions and microbial products that differ from those of the surrounding soil. A recent metagenomics study by Jeff Dangl (see the Unit 6 interview before Chapter 35) and his colleagues has revealed that the compositions of bacterial communities living endophytically and in the rhizosphere are not identical (**Figure 37.9**). A better understanding of the bacteria within and around roots could have profound agricultural benefits.

▼ **Figure 37.9**

Inquiry

How variable are the compositions of bacterial communities inside and outside of roots?

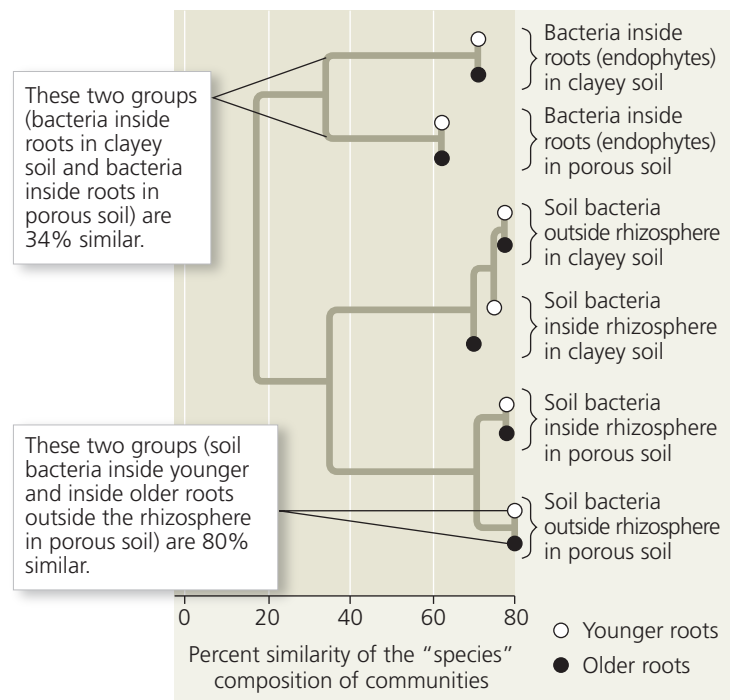
Experiment The bacterial communities found within and immediately outside of root systems are known to improve plant growth. In order to devise agricultural strategies to increase the benefits of these bacterial communities, it is necessary to determine how complex they are and what factors affect their composition. A problem inherent in studying these bacterial communities is that a handful of soil contains as many as 10,000 types of bacteria, more than all the bacterial species that have been described. One cannot simply culture each species and use a taxonomic key to identify them; a molecular approach is needed.



▲ Bacteria (green) on surface of root (fluorescent LM)

Jeffery Dangl (see the Unit 6 interview before Chapter 35) and his colleagues estimated the number of bacterial "species" in various samples using a technique called *metagenomics* (see Concept 21.1). The bacterial community samples they studied differed in location (endophytic, rhizospheric, or outside the rhizosphere), soil type (clayey or porous), and the developmental stage of the root system with which they were associated (old or young). The DNA from each sample was purified, and the polymerase chain reaction (PCR) was used to amplify the DNA that codes for the 16S ribosomal RNA subunits. Many thousands of DNA sequence variations were found in each sample. The researchers then lumped the sequences that were more than 97% identical into "taxonomic units" or "species." (The word *species* is in quotation marks because "two organisms having a single gene that is more than 97% identical" is not explicit in any definition of species.) Having established the types of "species" in each community, the researchers constructed a tree diagram showing the percent of bacterial "species" that were found in common in each community.

Results This tree diagram breaks down the relatedness of bacterial communities into finer and finer levels of detail. The two explanatory labels give examples of how to interpret the diagram.



Conclusion The "species" composition of the bacterial communities varied markedly according to the location inside the root versus outside the root and according to soil type.

INTERPRET THE DATA (a) Which of the three community locations was least like the other two? (b) Rank the three variables (community location, developmental stage of roots, and soil type) in terms of how strongly they affect the "species" composition of the bacterial communities.

Data from D.S. Lundberg et al., Defining the core *Arabidopsis thaliana* root microbiome, *Nature* 488:86–94 (2012).

Bacteria in the Nitrogen Cycle

Plants have mutualistic relationships with several groups of bacteria that help make nitrogen more available. From a global perspective, no mineral nutrient is more limiting to plant growth than nitrogen, which is required in large amounts for synthesizing proteins and nucleic acids. Here we focus on processes leading directly to nitrogen assimilation by plants.

Unlike other soil minerals, ammonium ions (NH_4^+) and nitrate ions (NO_3^-)—the forms of nitrogen that plants can use—are not derived from the weathering of rocks. Although lightning produces small amounts of NO_3^- that get carried to the soil in rain, most soil nitrogen comes from the activity of bacteria (**Figure 37.10**).

The **nitrogen cycle**, also discussed in Figure 55.14, describes transformations of nitrogen and nitrogenous compounds in nature. When a plant or animal dies, or an animal expels waste, the initial form of nitrogen is organic. Decomposers convert the organic nitrogen within the remains back into ammonium (NH_4^+), a process called **ammonification**. Other sources of soil NH_4^+ are *nitrogen-fixing bacteria* that convert gaseous nitrogen (N_2) to NH_3 , which then picks up another H^+ in the soil solution, forming NH_4^+ .

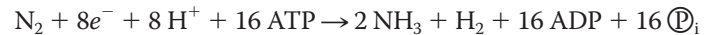
In addition to NH_4^+ , plants can also acquire nitrogen in the form of nitrate (NO_3^-). Soil NO_3^- is largely formed by a two-step process called **nitrification**, which consists of the oxidation of ammonia (NH_3) to nitrite (NO_2^-), followed by oxidation of NO_2^- to NO_3^- . Different types of *nitrifying bacteria* mediate each step. After the roots absorb NO_3^- , a plant enzyme reduces it back to NH_4^+ , which other enzymes incorporate into amino acids and other organic compounds. Most plant species export nitrogen from roots to shoots via

the xylem as NO_3^- or organic compounds synthesized in the roots. Some soil nitrogen is lost, particularly in anaerobic soils, when denitrifying bacteria convert NO_3^- to N_2 , which diffuses into the atmosphere.

Nitrogen-Fixing Bacteria: A Closer Look

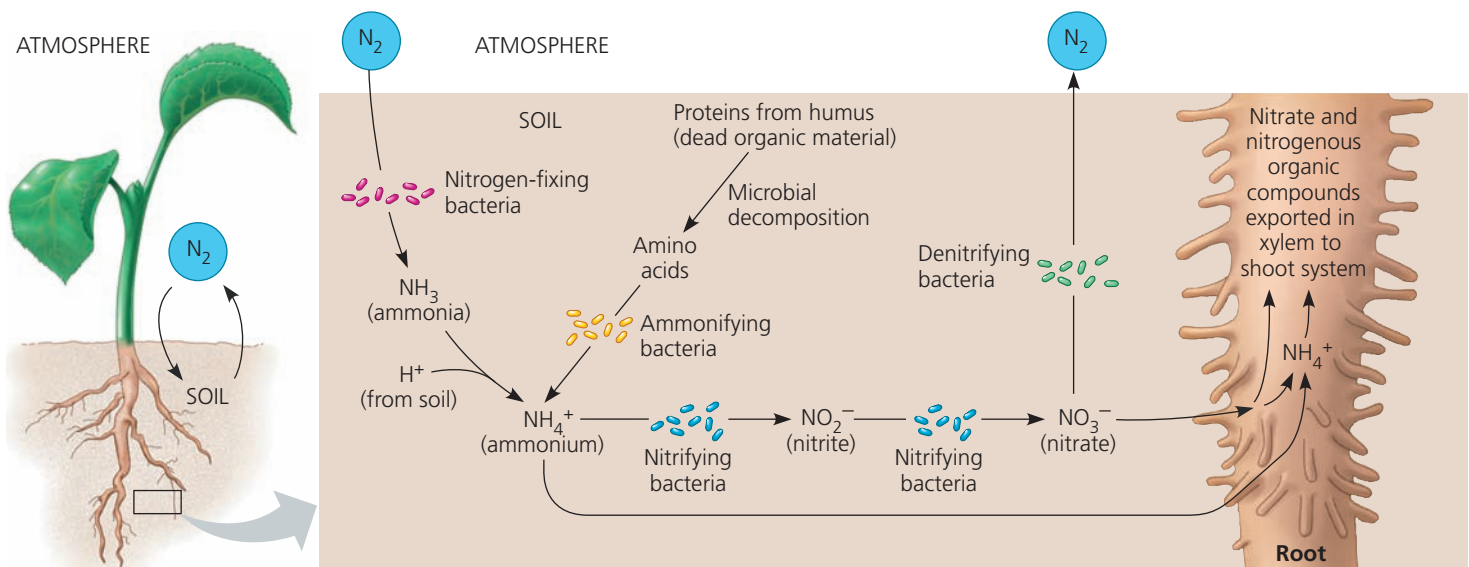
Although Earth's atmosphere is 79% nitrogen, plants cannot use free gaseous nitrogen (N_2) because there is a triple bond between the two nitrogen atoms, making the molecule almost inert. For atmospheric N_2 to be of use to plants, it must be reduced to NH_3 by a process known as **nitrogen fixation**. All nitrogen-fixing organisms are bacteria. Some nitrogen-fixing bacteria are free-living in the soil (see Figure 37.10), whereas others are endophytic. Still others, particularly members of the genus *Rhizobium*, form efficient and intimate associations with the roots of legumes (such as peas, soybeans, alfalfa, and peanuts), altering the structure of the hosts' roots markedly, as will be discussed shortly.

The multistep conversion of N_2 to NH_3 by nitrogen fixation can be summarized as follows:



The reaction is driven by the enzyme complex *nitrogenase*. Because the process of nitrogen fixation requires 16 ATP molecules for every 2 NH_3 molecules synthesized, nitrogen-fixing bacteria require a rich supply of carbohydrates from decaying material, root secretions, or (in the case of the *Rhizobium* bacteria) the vascular tissue of roots.

The specialized mutualism between *Rhizobium* bacteria and legume roots involves dramatic changes in root structure. Along a legume's roots are swellings called **nodules**,



▲ Figure 37.10 The roles of soil bacteria in the nitrogen nutrition of plants. Ammonium is made available to plants by two types of soil bacteria: those that fix atmospheric N_2

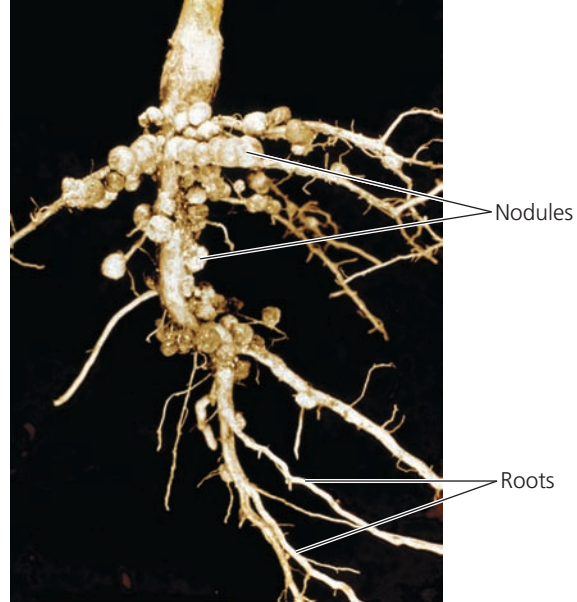
(nitrogen-fixing bacteria) and those that decompose organic material (ammonifying bacteria). Although plants absorb some ammonium from the soil, they absorb mainly nitrate, which is

produced from ammonium by nitrifying bacteria. Plants reduce nitrate back to ammonium before incorporating the nitrogen into organic compounds.

composed of plant cells “infected” by *Rhizobium* (“root living”) bacteria (**Figure 37.11**). Inside each nodule, *Rhizobium* bacteria assume a form called **bacteroids**, which are contained within vesicles formed in the root cells. Legume-*Rhizobium* relationships generate more usable nitrogen for plants than all industrial fertilizers used today, and the mutualism provides the right amount of nitrogen at the right time at virtually no cost to the farmer.

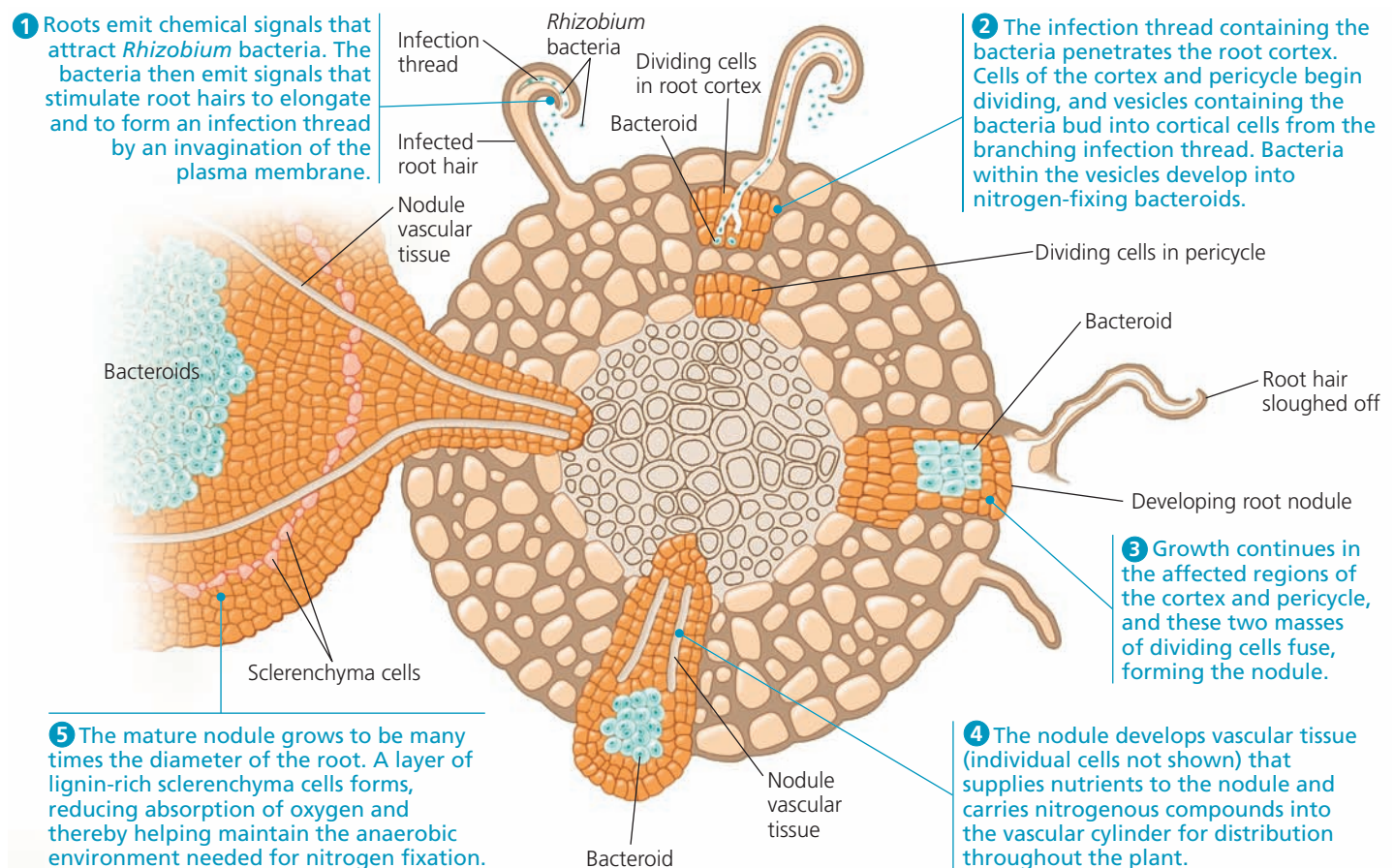
The location of the bacteroids inside living, nonphotosynthetic cells is conducive to nitrogen fixation, which requires an anaerobic environment. Lignified external layers of root nodules also limit gas exchange. Some root nodules appear reddish because of a molecule called leghemoglobin (*leg-* for “legume”), an iron-containing protein that binds reversibly to oxygen (similar to the hemoglobin in human red blood cells). This protein is an oxygen “buffer,” reducing the concentration of free oxygen and thereby providing an anaerobic environment for nitrogen fixation while regulating the oxygen supply for the intense cellular respiration required to produce ATP for nitrogen fixation.

Each legume species is associated with a particular strain of *Rhizobium*. **Figure 37.12** describes how a root nodule develops after bacteria enter through an “infection thread.”



▲ **Figure 37.11 Root nodules on a legume.** The spherical structures along this soybean root system are nodules containing *Rhizobium* bacteria. The bacteria fix nitrogen and obtain photosynthetic products supplied by the plant.

? How is the relationship between legume plants and *Rhizobium* bacteria mutualistic?



▲ **Figure 37.12 Development of a soybean root nodule.**

? What plant tissue systems are modified by root nodule formation?

The symbiotic relationship between a legume and nitrogen-fixing bacteria is mutualistic in that the bacteria supply the host plant with fixed nitrogen while the plant provides the bacteria with carbohydrates and other organic compounds. The root nodules use most of the ammonium produced to make amino acids, which are then transported up to the shoot through the xylem.

How does a legume species recognize a certain strain of *Rhizobium* among the many bacterial strains in the soil? And how does an encounter with that specific *Rhizobium* strain lead to development of a nodule? These two questions have led researchers to uncover a chemical dialogue between the bacteria and the root. Each partner responds to chemical signals from the other by expressing certain genes whose products contribute to nodule formation. By understanding the molecular biology underlying the formation of root nodules, researchers hope to learn how to induce *Rhizobium* uptake and nodule formation in crop plants that do not normally form such nitrogen-fixing mutualistic relationships.

Nitrogen Fixation and Agriculture

The agricultural benefits of mutualistic nitrogen fixation underlie most types of **crop rotation**. In this practice, a non-legume such as maize is planted one year, and the following year alfalfa or some other legume is planted to restore the concentration of fixed nitrogen in the soil. To ensure that the legume encounters its specific *Rhizobium* strain, the seeds are exposed to bacteria before sowing. Instead of being harvested, the legume crop is often plowed under so that it will decompose as “green manure,” reducing the need for manufactured fertilizers.

Many plant families besides legumes include species that benefit from mutualistic nitrogen fixation. For example, alder trees and certain tropical grasses host nitrogen-fixing actinomycete bacteria (see the gram-positive bacteria in Figure 27.16). Rice, a crop of great commercial importance, benefits indirectly from mutualistic nitrogen fixation. Rice farmers culture a free-floating aquatic fern, *Azolla*, which has mutualistic cyanobacteria that fix N_2 . The growing rice eventually shades and kills the *Azolla*, and decomposition of this nitrogen-rich organic material increases the paddy's fertility.

Fungi and Plant Nutrition

Certain species of soil fungi also form mutualistic relationships with roots and play a major role in plant nutrition. Some of these fungi are endophytic, but the most important relationships are **mycorrhizae** (“fungus roots”), the intimate mutualistic associations of roots and fungi (see Figure 31.14). The host plant provides the fungus with a steady supply of sugar. Meanwhile, the fungus increases the surface area for

water uptake and also supplies the plant with phosphate and other minerals absorbed from the soil. The fungi of mycorrhizae also secrete growth factors that stimulate roots to grow and branch, as well as antibiotics that help protect the plant from soil pathogens.

Mycorrhizae and Plant Evolution

EVOLUTION Mycorrhizae are not oddities; they are formed by most plant species. In fact, this plant-fungus mutualism might have been one of the evolutionary adaptations that helped plants initially colonize land (see Chapter 29). When the earliest plants, which evolved from green algae, began to invade the land 400 to 500 million years ago, they encountered a harsh environment. Although the soil contained mineral nutrients, it lacked organic matter. Therefore, rain probably quickly leached away many of the soluble mineral nutrients. The barren land, however, was also a place of opportunities because there was very little competition, and light and carbon dioxide were readily available. Neither the early land plants nor early land fungi, which evolved from an aquatic protist, were fully equipped to exploit the terrestrial environment. The early plants lacked the ability to extract essential nutrients from the soil, while the fungi were unable to manufacture carbohydrates. Instead of the fungi becoming parasitic on the rhizoids of the evolving plants (roots or root hairs had not yet evolved), the two organisms formed mycorrhizal associations, a mutualistic symbiosis that allowed both of them to exploit the terrestrial environment. Fossil evidence supports the idea that mycorrhizal associations occurred in the earliest land plants. The small minority of extant angiosperms that are nonmycorrhizal probably lost this ability through gene loss.

The Two Main Types of Mycorrhizae

One type of mycorrhizae—the **ectomycorrhizae**—form a dense sheath, or mantle of mycelia (mass of branching hyphae; see Chapter 31), over the *surface* of the root (**Figure 37.13a**). Fungal hyphae extend from the mantle into the soil, greatly increasing the surface area for water and mineral absorption. Hyphae also grow into the root cortex. These hyphae do not penetrate the root cells but form a network in the apoplast, or extracellular space, that facilitates nutrient exchange between the fungus and the plant. Compared with “uninfected” roots, ectomycorrhizae are generally thicker, shorter, and more branched. They typically do not form root hairs, which would be superfluous given the extensive surface area of the fungal mycelium. About 10% of plant families have species that form ectomycorrhizae, and the vast majority of these species are woody, including members of the pine, oak, birch, and eucalyptus families.

Unlike ectomycorrhizae, **arbuscular mycorrhizae** do not form a dense mantle ensheathing the root (**Figure 37.13b**).

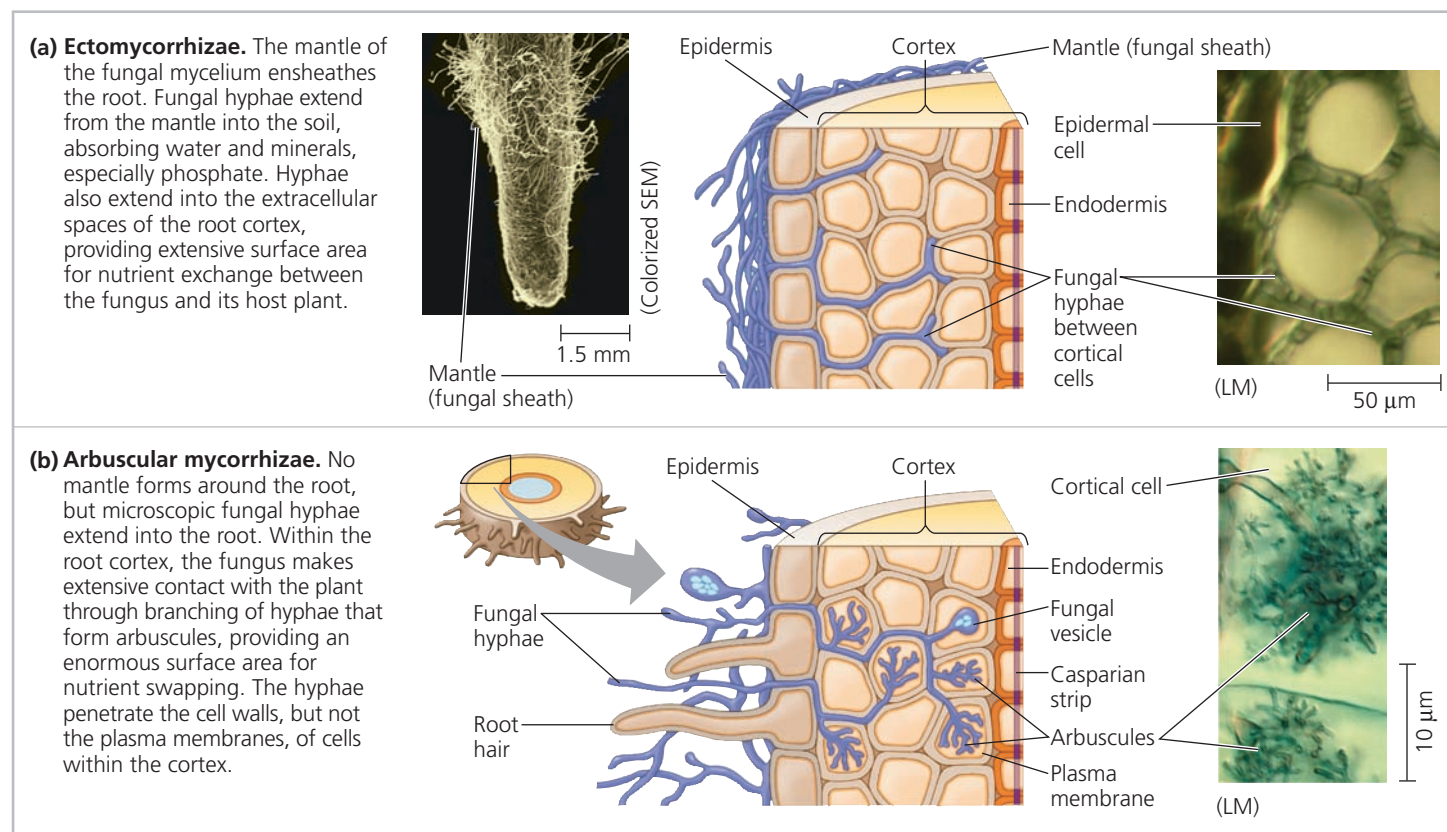
Arbuscular mycorrhizae are much more common than ectomycorrhizae and are found in over 85% of plant species. Among those species are most crop plants, including grains and legumes.

To the unaided eye, arbuscular mycorrhizae look like “normal” roots with root hairs, but a microscope reveals the enormous extent of the mutualistic relationship. Arbuscular mycorrhizal associations start when microscopic soil hyphae respond to the presence of a root by growing toward it, establishing contact, and growing along its surface. The hyphae penetrate between epidermal cells and then enter the root cortex. These hyphae digest small patches of the cortical cell walls, but they do not actually pierce the plasma membrane and enter the cytoplasm. Instead, a hypha grows into a tube formed by invagination of the root cell’s membrane. The process of invagination is analogous to poking a finger gently into a balloon without popping it; your finger is like the fungal hypha, and the balloon skin is like the root cell’s membrane. After the fungal hyphae have penetrated in this way, some of them branch densely, forming structures called arbuscules (“little trees”), which are important sites of nutrient transfer between the fungus and the plant. Within the hyphae themselves, oval vesicles may form, possibly serving as food storage sites for the fungus.

Agricultural and Ecological Importance of Mycorrhizae

Roots can form mycorrhizal symbioses only if exposed to the appropriate species of fungus. In most ecosystems, these fungi are present in the soil, and seedlings develop mycorrhizae. But if seeds are collected in one environment and planted in foreign soil, the plants may show signs of malnutrition (particularly phosphorus deficiency), resulting from the absence of fungal partners. Treating seeds with spores of mycorrhizal fungi can sometimes help seedlings to form mycorrhizae and improve crop yield.

Mycorrhizal associations are also important in understanding ecological relationships. Arbuscular mycorrhizae fungi exhibit little host specificity; a single fungus may form a shared mycorrhizal network with several plants, even plants of different species. Mycorrhizal networks in a plant community may benefit one plant species more than another. Another example of how mycorrhizae may affect the structures of plant communities comes from studies of exotic invasive plant species. Garlic mustard (*Alliaria petiolata*), an exotic European species that has invaded woodlands throughout the eastern United States, does not form mycorrhizae but hinders the growth of other plant species by preventing the growth of arbuscular mycorrhizal fungi.



▲ **Figure 37.13** Mycorrhizae.

Exploring Unusual Nutritional Adaptations in Plants

Epiphytes

An **epiphyte** (from the Greek *epi*, upon, and *phyton*, plant) is a plant that grows on another plant. Epiphytes produce and gather their own nutrients; they do not tap into their hosts for sustenance. Usually anchored to the branches or trunks of living trees, epiphytes absorb water and minerals from rain, mostly through leaves rather than roots. Some examples are staghorn ferns, bromeliads, and many orchids, including the vanilla plant.

► Staghorn fern, an epiphyte

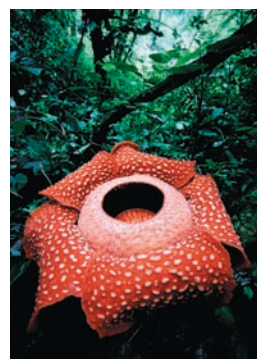


Parasitic Plants



Unlike epiphytes, parasitic plants absorb water, minerals, and sometimes products of photosynthesis from their living hosts. Many species have roots that function as haustoria, nutrient-absorbing projections that tap into the host plant. Some parasitic species, such as *Rafflesia arnoldii*, lack chlorophyll entirely, whereas others, such as mistletoe (genus *Phoradendron*), are photosynthetic. Still others, such as Indian pipe (*Monotropa uniflora*), absorb nutrients from the hyphae of mycorrhizae associated with other plants.

◄ Mistletoe, a photosynthetic parasite



▲ *Rafflesia*, a nonphotosynthetic parasite



▲ Indian pipe, a nonphotosynthetic parasite of mycorrhizae

Carnivorous Plants

Carnivorous plants are photosynthetic but supplement their mineral diet by capturing insects and other small animals. They live in acid bogs and other habitats where soils are poor in nitrogen and other minerals. Pitcher plants such as *Nepenthes* and *Sarracenia* have water-filled funnels into which prey slip and drown, eventually to be digested by enzymes. Sundews (genus *Drosera*) exude a sticky fluid from tentacle-like glands on highly modified leaves. Stalked glands secrete sweet mucilage that attracts and ensnares insects, and they also release digestive enzymes. Other glands then absorb the nutrient "soup." The highly modified leaves of Venus flytrap (*Dionaea muscipula*) close quickly but partially when a prey hits two trigger hairs in rapid enough succession. Smaller insects can escape, but larger ones are trapped by the teeth lining the margins of the lobes. Excitation by the prey causes the trap to narrow more and digestive enzymes to be released.



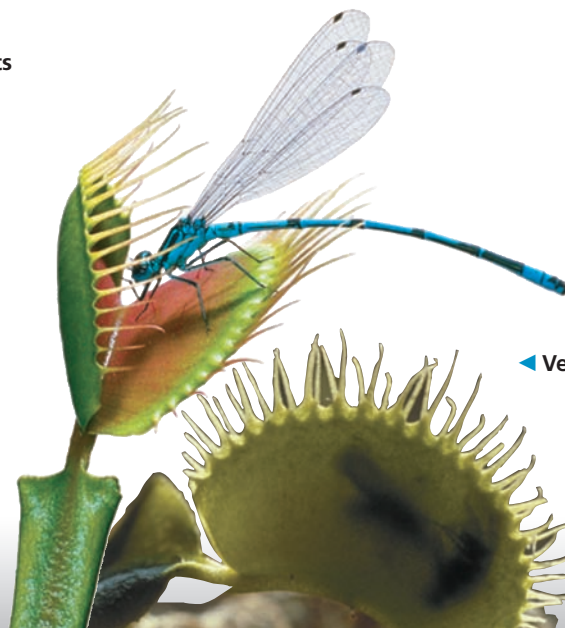
◄ Pitcher plants



▲ Sundew



◄ Venus flytraps



Epiphytes, Parasitic Plants, and Carnivorous Plants

Almost all plant species have mutualistic relationships with soil fungi, bacteria, or both. Some plant species, including epiphytes, parasites, and carnivores, have unusual adaptations that facilitate exploiting other organisms (Figure 37.14).

A recent study suggests that such behaviors may be the norm. Chanyarat Paungfoo-Lonhienne and her colleagues at the University of Queensland in Australia have provided evidence that *Arabidopsis* and tomato can take up bacteria and yeast into their roots and digest them. This pioneering research suggests that carnivory by plants may not be an adaptation limited to only a handful of odd species such as *Genlisea* (see Figure 37.1), but that many plant species might engage in a limited amount of heterotrophy.

CONCEPT CHECK 37.3

1. Why is the study of the rhizosphere critical to understanding plant nutrition?
2. How do soil bacteria and mycorrhizae contribute to plant nutrition?
3. **MAKE CONNECTIONS** What is a general term used to describe the strategy of using photosynthesis *and* heterotrophy for nutrition (see Chapter 28)? What is a well-known example of a class of protists that uses this strategy?
4. **WHAT IF?** A peanut farmer finds that the older leaves of his plants are turning yellow following a long period of wet weather. Suggest a reason why.

For suggested answers, see Appendix A.

37 Chapter Review

SUMMARY OF KEY CONCEPTS

CONCEPT 37.1

Soil contains a living, complex ecosystem (pp. 800–803)

- Soil particles of various sizes derived from the breakdown of rock are found in soil. Soil particle size affects the availability of water, oxygen, and minerals in the soil.
- A soil's composition refers to its inorganic and organic components. **Topsoil** is a complex ecosystem teeming with bacteria, fungi, protists, animals, and the roots of plants.
- Some agricultural practices can deplete the mineral content of soil, tax water reserves, and promote erosion. The goal of soil conservation is to minimize this damage.

? How is soil a complex ecosystem?

CONCEPT 37.2

Plants require essential elements to complete their life cycle (pp. 803–806)

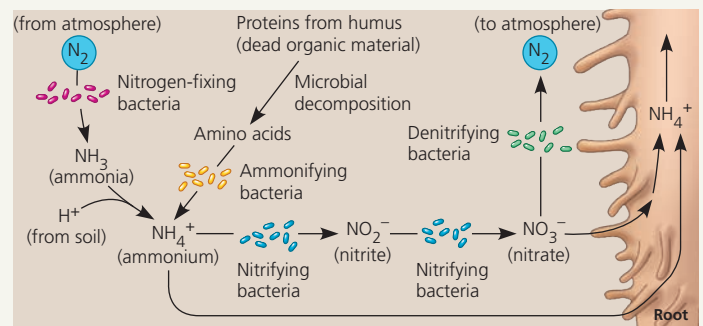
- **Macronutrients**, elements required in relatively large amounts, include carbon, oxygen, hydrogen, nitrogen, and other major ingredients of organic compounds. **Micronutrients**, elements required in very small amounts, typically have catalytic functions as cofactors of enzymes.
- Deficiency of a mobile nutrient usually affects older organs more than younger ones; the reverse is true for nutrients that are less mobile within a plant. Macronutrient deficiencies are most common, particularly deficiencies of nitrogen, phosphorus, and potassium.
- Rather than tailoring the soil to match the plant, genetic engineers are tailoring the plant to match the soil.

? Do plants need soil to grow? Explain.

CONCEPT 37.3

Plant nutrition often involves relationships with other organisms (pp. 806–813)

- **Rhizobacteria** derive their energy from the **rhizosphere**, a microbe-enriched ecosystem intimately associated with roots. Plant secretions support the energy needs of the rhizosphere. Some rhizobacteria produce antibiotics, whereas others make nutrients more available for plants. Most are free-living, but some live inside plants. Plants satisfy most of their huge needs for nitrogen from the bacterial decomposition of **humus** and the fixation of gaseous nitrogen.



Nitrogen-fixing bacteria convert atmospheric N_2 to nitrogenous minerals that plants can absorb as a nitrogen source for organic synthesis. The most efficient mutualism between plants and nitrogen-fixing bacteria occurs in the **nodules** that are formed by *Rhizobium* bacteria growing in the roots of legumes. These bacteria obtain sugar from the plant and supply the plant with fixed nitrogen. In agriculture, legume crops are rotated with other crops to restore nitrogen to the soil.

- **Mycorrhizae** are mutualistic associations of fungi and roots. The fungal hyphae of mycorrhizae absorb water and minerals, which they supply to their plant hosts.
- **Epiphytes** grow on the surfaces of other plants but acquire water and minerals from rain. Parasitic plants absorb nutrients from host plants. Carnivorous plants supplement their mineral nutrition by digesting animals.

? Do all plants gain energy directly from photosynthesis? Explain.

TEST YOUR UNDERSTANDING

LEVEL 1: KNOWLEDGE/COMPREHENSION

- The inorganic nutrient most often lacking in crops is
 - carbon.
 - nitrogen.
 - phosphorus.
 - potassium.
- Micronutrients are needed in very small amounts because
 - most of them are mobile in the plant.
 - most serve mainly as cofactors of enzymes.
 - most are supplied in large enough quantities in seeds.
 - they play only a minor role in the growth and health of the plant.
- Mycorrhizae enhance plant nutrition mainly by
 - absorbing water and minerals through the fungal hyphae.
 - providing sugar to root cells, which have no chloroplasts.
 - converting atmospheric nitrogen to ammonia.
 - enabling the roots to parasitize neighboring plants.
- Epiphytes are
 - fungi that attack plants.
 - fungi that form mutualistic associations with roots.
 - nonphotosynthetic parasitic plants.
 - plants that grow on other plants.
- Some of the problems associated with intensive irrigation include all of the following except
 - soil salinization.
 - overfertilization.
 - land subsidence.
 - aquifer depletion.

LEVEL 2: APPLICATION/ANALYSIS

- A mineral deficiency is likely to affect older leaves more than younger leaves if
 - the mineral is a micronutrient.
 - the mineral is very mobile within the plant.
 - the mineral is required for chlorophyll synthesis.
 - the mineral is a macronutrient.
- The greatest difference in health between two groups of plants of the same species, one group with mycorrhizae and one group without mycorrhizae, would be in an environment
 - where nitrogen-fixing bacteria are abundant.
 - that has soil with poor drainage.
 - that has hot summers and cold winters.
 - in which the soil is relatively deficient in mineral nutrients.
- Two groups of tomatoes were grown under laboratory conditions, one with humus added to the soil and one a control without humus. The leaves of the plants grown without humus were yellowish (less green) compared with those of the plants grown in humus-enriched soil. The best explanation is that
 - the healthy plants used the food in the decomposing leaves of the humus for energy to make chlorophyll.
 - the humus made the soil more loosely packed, so water penetrated more easily to the roots.
 - the humus contained minerals such as magnesium and iron needed for the synthesis of chlorophyll.
 - the heat released by the decomposing leaves of the humus caused more rapid growth and chlorophyll synthesis.

- The specific relationship between a legume and its mutualistic *Rhizobium* strain probably depends on
 - each legume having a chemical dialogue with a fungus.
 - each *Rhizobium* strain having a form of nitrogenase that works only in the appropriate legume host.
 - each legume being found where the soil has only the *Rhizobium* specific to that legume.
 - specific recognition between chemical signals and signal receptors of the *Rhizobium* strain and legume species.
- DRAW IT** Draw a simple sketch of cation exchange, showing a root hair, a soil particle with anions, and a hydrogen ion displacing a mineral cation.

LEVEL 3: SYNTHESIS/EVALUATION

11. EVOLUTION CONNECTION

Imagine taking the plant out of the picture in Figure 37.10. Write a paragraph explaining how soil bacteria could sustain the recycling of nitrogen *before* land plants evolved.

12. SCIENTIFIC INQUIRY

Acid precipitation has an abnormally high concentration of hydrogen ions (H^+). One effect of acid precipitation is to deplete the soil of nutrients such as calcium (Ca^{2+}), potassium (K^+), and magnesium (Mg^{2+}). Suggest a hypothesis to explain how acid precipitation washes these nutrients from the soil. How might you test your hypothesis?

13. SCIENCE, TECHNOLOGY, AND SOCIETY

In many countries, irrigation is depleting aquifers to such an extent that land is subsiding, harvests are decreasing, and it is becoming necessary to drill wells deeper. In many cases, the withdrawal of groundwater has now greatly surpassed the aquifers' rates of natural recharge. Discuss the possible consequences of this trend. What can society and science do to help alleviate this growing problem?

14. WRITE ABOUT A THEME: INTERACTIONS

The soil in which plants grow teems with organisms from every taxonomic kingdom. In a short essay (100–150 words), discuss examples of how the mutualistic interactions of plants with bacteria, fungi, and animals improve plant nutrition.

15. SYNTHESIZE YOUR KNOWLEDGE



Making a footprint in the soil seems like an insignificant event. In a short essay (100–150 words), explain how a footprint would affect the properties of the soil and how these changes would affect soil organisms and the emergence of seedlings.

For suggested answers, see Appendix A.

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